Unit 2, Lesson 1: Equivalent Conversions: Fractions ↔ Decimals; Decimals ↔ Percentages

Benchmark

MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.

Learning Targets

- ☐ I can convert between fractions and decimals.
- ☐ I can convert between percentages and decimals.

2.1.1 Warm-Up: Self-Evaluation

1. For each problem, rate your level of confidence in your ability to answer each question, but don't actually answer them.

   Use a scale from 1 to 5, where 1 means you do not feel confident about how to answer the question, and 5 means you feel very confident.

Problem	Confidence Level
Rewrite $2\frac{2}{5}$ in its equivalent decimal form.	
Rewrite 0.625 as a fraction with the same value.	
Rewrite 0.9 as an equivalent percentage.	
Rewrite 75% in its equivalent decimal form.	

2.1.2 Guided Instruction: Fraction and Decimal Conversions

Decimal Place Value Chart

Thousands	Hundreds	Tens	Ones	Decimal Point	Tenths	Hundredths	Thousandths

The place value of a number provides the numerical value that a digit has by virtue of its position in a number.

Example: In 278.493
the 8 is in the **ones** place
the 4 is in the **tenths** place
the 9 is in the **hundredths** place
the 3 is in the **thousandths** place

1. Use the original number to fill in the missing digits and place values.

Original Number	Digit	Place Value
422.54	5	Tenths
5.276		Ones
12.749	4	
3.826		Thousandths

2. Consider the number 0.08.

   a. In the number 0.08, what is the place value of the final digit?

   b. What is the written description of the fraction $\frac{8}{100}$?

   c. Discuss with your partner any similarities or differences between the decimal 0.08 and the fraction $\frac{8}{100}$.

When rewriting a decimal that is greater than 0 but less than 1 as an equivalent fraction, use the place value of the final digit to determine the denominator of the fraction.

3. In the decimal 0.7, the final digit is in the

- ○ tenths
- ○ hundredths
- ○ thousandths

place, so

- ○ $\frac{7}{10}$
- ○ $\frac{7}{100}$
- ○ $\frac{7}{1000}$

is the equivalent fraction form.

4. In the decimal 0.037, the final digit is in the

- ○ tenths
- ○ hundredths
- ○ thousandths

place, so

- ○ $\frac{37}{10}$
- ○ $\frac{37}{100}$
- ○ $\frac{37}{1000}$

is the equivalent fraction form.

A fraction can be converted into its equivalent decimal form by dividing the ***numerator*** by the ***denominator***.

5. To convert the fraction $\frac{1}{4}$ to a decimal, divide 1 by 4 by completing the diagram below.

$$4 \overline{)\ 1.00}$$

Number of ones groups of 4	−	______
Amount that remains		______
Number of tenths groups of 4	−	______
Amount that remains		______
Number of hundredths groups of 4	−	______
Amount that remains		______

2.1.3 Guided Practice: Fraction and Decimal Conversions

1. Complete the table by converting each decimal to an equivalent fraction. If possible, write the equivalent fraction so that the numerator and denominator have no common factors.

Decimal Form	Fraction Form
0.08	$\frac{2}{25}$
0.27	
2.2	
0.005	

2. Complete the table by rewriting each fraction as its equivalent decimal.

Fraction Form	Your Work	Decimal Form
$\frac{1}{5}$		
$\frac{5}{8}$		
$\frac{7}{200}$		

2.1.4 Guided Instruction: Decimal and Percentage Conversions

A ***percentage*** is a ratio expressed as a fraction with a denominator of 100.

Percentages can also be written as fractions with a denominator of 100, or as a decimal.

1. Write 66% as a fraction and as a decimal.

Percent	Fraction	Decimal
66%	$\frac{\quad}{100}$	

To convert a decimal to a percent, multiply the decimal by 100.

2. Convert the decimal 0.16 to a percent.

$$0.16 \times \_\_\_\_\_\_$$
$$\_\_\_\_\_\_\%$$

To convert a percent to a decimal, divide the percent by 100 or multiply by $\frac{1}{100}$.

3. Convert 16.2% to a decimal.

$$16.2 \div \_\_\_\_\_\_ \text{ or } 16.2 \cdot \_\_\_\_\_\_$$
$$\_\_\_\_\_\_\_\_\_\_$$

2.1.5 Guided Practice: Decimal and Percentage Conversions

1. Complete the table by converting each decimal to an equivalent percentage.

Decimal Form	Percent Form
0.08	8%
0.27	
2.2	
0.005	

2. Rewrite each percentage into its equivalent decimal form to complete the table.

Percent Form	Decimal Form
72%	0.72
12.5%	
0.7%	
214%	

2.1.6 Your Turn!

1. Rewrite each percentage as a decimal.

   a. 125.6% b. 3%

2. Convert each decimal to an equivalent percentage.

   a. 0.008 b. 4.71

3. Rewrite each fraction as a decimal.

   a. $\frac{1}{2}$ b. $2\frac{1}{10}$

4. Rewrite each decimal as an equivalent fraction. Simplify where applicable.

   a. 0.94 b. 1.7

2.1.7 Wrap-Up: Ticket Out the Door

1. Circle the expression that does not belong.

$\frac{2}{5}$	0.4
40%	$\frac{4}{8}$

2. Explain your reasoning for your choice.